

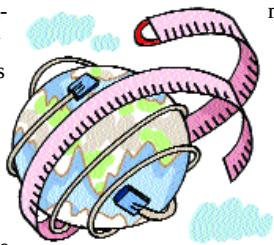
Global measures

Miles of fibre-optic cable may answer a question first tackled by the ancient Greeks: How big is the Earth?

In a recent paper, Youngstown State University physics teacher Michael Crescimanno describes how computer signals travelling under-sea through buried fibre-optic cables can be used to quickly and easily estimate the size of the planet. "The

traceroute utility on any computer connected to the Internet can be used to record the round-trip time for small Internet packets between major Internet traffic hubs,"

he explains. Crescimanno's students have developed the traceroute experiment as an inexpensive way to involve introductory physics students in hands-on use of scientific notation



Never buy a game again

Yahoo!, always a path-setter of one kind or another, now brings Yahoo! Games, which serves 4.5 billion minutes of game time to 8 million players each month. The site peaks each day with about 1,50,000 simultaneous players and now Yahoo! hopes to work around the troubles of renting games by doing it online. On September 23, they launched their Yahoo!



Games On Demand service that allows broadband PC users to rent titles at very competitive prices. Yahoo! has overcome security issues by using a custom client for playing the games, so you never

and to teach them about systematics in data.

Unfortunately, the Earth's topography hinders perfectly accurate measurements because cables are not laid precisely along great circles. Crescimanno has attempted to find some means of error-correction by comparing road and straight-line distances between eight Ohio cities. Traceroute-based measurements of the Earth's radius yield errors similar to his findings of a 20 per cent difference between the Ohio cities.

download the whole package, and some of the biggest game publishers have teamed up with them.

Currently there are 40 titles available for rent at <http://gamesondemand.yahoo.com>, and all you need to do is login to the Yahoo! Games On Demand service and pay for a rental. A three day rental for one game costs \$4.95 (approx Rs 250). You can also get a package deal at higher prices, for as long as a month.

IDF 2002

While all of America was mourning on September 11, San Jose, California was buzzing with activity as the Intel Developer Forum (IDF) showed off some of the latest technologies and developments from September 9 to 12. Frank Spindler, vice president of Corporate Technology Group, outlined the key subjects of the forum. The schedule was tightly packed with briefings, conferences, demonstration of products and technologies and special parties, but that didn't stop Intel from throwing in tennis and game tables for people to relax. All in all, the forum was a technological paradise. You can visit the official site for the forum at www.intel.com/idf/us/fall2002/index.htm.

snapshot

IT spending worldwide expected to total **\$2.3 trillion** in 2002 and to rise to **\$2.4 trillion** in 2003

Source: Gartner Dataquest

Apple teaming up with IBM

IBM Microelectronics is working with Apple on a 64-bit PowerPC processor for use in Apple's high-end desktops and servers. Apple is testing the CPU, dubbed the GigaProcessor Ultralite (GPUL), on Mac OS X hardware and making sure that the processor complies with a new bus architecture for future Macs. In addition, IBM plans to offer the processor as the centrepiece of future Linux-based systems. GPUL, produced on 130 nm technology,

is expected to run in the 1.4-2.0 GHz range and will double the Mac performance. While the power-hungry

Power4 is designed for servers, GPUL is reportedly cooler and more compact, comparing in size to Intel's



Celeron. However, GPUL will inherit many Power4 performance advantages, such as being able to perform more instructions per clock cycle

than current PowerPC chips. GPUL will be a multi-core chip, with two or four processors in one package. Having the processors closer together and sharing the same cache will make for faster multiprocessing environments.

However, it is not yet clear whether applications would need to be rewritten to take advantage of this architecture.

Benchmarks and applications tests show that a 1 GHz

GPUL processor provides double the performance of a 1 GHz Motorola PowerPC G4 processor. GPUL will support Vector/SIMD Multimedia Extensions (VMX), a group of 162 instructions that speed data processing and algorithm-intensive tasks, such as multimedia creation and display. VMX has been designed to be compatible with AltiVec multimedia extensions currently used in the PowerPC G4.



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